USING TENSOR FACTORIZATION TO CAPTURE SPATIOTEMPORAL PATTERNS FOR SYNCHRONIZATION PREDICTION

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ABSTRACT

The clock synchronization problem has been studied for decades, and it has far-fetched applications in robotics, distributed computing systems, formation of public opinion, and even development of cancer cells. However, the task of predicting whether a system of coupled oscillators on an underlying graph would eventually synchronize or not is an analytically intractable one. While recent studies have tried to tackle this issue with the use of machine learning models, there remains the seek for an interpretable framework for this problem that can even be used to answer 'how' the model makes its predictions and 'what' features the model considers as important for the same. Leveraging the powerful feature representation learning capabilities of techniques like matrix factorization and Tensor CP-Decomposition, we want to propose an interpretable framework for the prediction of synchronization in coupled oscillators specifically making use of nonnegativity constraints on our features. We believe that having an interpretable framework for this problem which also has good prediction accuracy compared to the baseline, would prove to be crucial to the advancement of our understanding of oscillator theory as well as allow for wide adoption of machine learning based techniques for more important applications of coupled oscillators.

Introduction

Theoretical understanding of the large-scale behavior of any complex systems consisting of locally interacting dynamic agents has been an important subject of research in various areas of science and mathematics for decades.

In particular, one of the most widely studied of such collective behaviors has been the long-term global synchronization of coupled oscillators (e.g., circadian pacemakers, blinking fireflies, and BZ chemical oscillators) [Strogatz (2000), Acebrón et al. (2005)]. The analysis of the long-term synchronization behaviors in such kind of networked systems can be used to model many common but important phenomena that have gained a lot of importance in recent years, such as the formation of public opinion, the development of cancer cells, the outbreak of epidemics, as well as collective computation in distributed computing systems. This problem is further closely linked to that of clock synchronization, which is essential in establishing shared notions of time in distributed systems and has enjoyed fruitful applications in many areas including designing distributed algorithms, wildfire monitoring, electric power networks, robotic vehicle networks,

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large-scale information fusion, and wireless sensor networks [Dorfler and Bullo (2012), Nair and Leonard (2007), Pagliari and Scaglione (2010)]. Synchronization occurs when the oscillators spontaneously lock into a common frequency or phase within the coupled dynamical system.

The problem of predicting whether a given system of coupled oscillators on an underlying arbitrary graph structure will eventually synchronize or not, although relevant in a variety of fields, is analytically intractable. Moreover, in spite of having the knowledge about several sufficient conditions for model parameters (e.g., large coupling strength of the oscillators) or initial configuration [Lyu (2018)] (e.g., phase concentration at the start) as well as graph topology [Eom et al. (2016), Boccaletti et al. (2006), Chowdhury et al. (2020)] (e.g., complete graphs or trees), it is elusive to obtain analytical or asymptotic solutions to the synchronization prediction problems. In fact, due to the non-linear local interactions of oscillators, the complexity of this problem increases exponentially as the graph topology becomes more complex and number of nodes in the graphs increases. An example of this is that there exist more than 10^9 non-isomorphic simple graphs for just 11 nodes of complexity [McKay].

Recent studies have made attempts to approach this clock synchronization problem using traditional machine learning methods [Bassi et al. (2022), Fan et al. (2021), Biccari and Zuazua (2020)], and they show a promising direction to approach the synchronization problem as they achieve very high accuracy compared to what we expect from theoretical baselines, and at the same time achieve this accuracy in a reasonable number of iterations. However, a key aspect of this problem that hasn't been studied enough is the "interpretability" of *how* a machine learning model is able to make these predictions with such high accuracy and *what* are the main features that these models consider relevant while making these predictions. This puts the current methods into the category of a 'blackbox' as with most machine learning based models, where although great results can be achieved, the entire framework as a whole is not interpretable. Our work makes an attempt to tackle the lack of emphasis placed by prior studies on having an interpretable framework for the prediction of synchronization.

In particular, we will leverage the utility of feature representation learning algorithms like nonnegative matrix factorization [Lee and Seung (1999)], (online) supervised dictionary learning [Mairal et al. (2008), Mairal et al. (2010)], and tensor decomposition [Tucker (1966), Harshman et al. (1970)] to learn interpretable features from a tensor representation of these graph-dynamic pairs (See 'Methods' section), and use these features for the purpose of prediction of synchronization. This is crucial since being able to have an interpretable framework that would also perform well would provide useful insights to the coupled oscillator community. Further, it will allow the industry to use these machine learning based methods for predicting synchronization in important and yet sensitive applications like studying the development of cancer cells since the community will be aware of how the model has arrived at the prediction.

Statement of the problem

A graph $G = (V, E)$ consists of sets V of vertices (or nodes) and E of edges. Let Ω denote the *phase space* of each node, which may be taken to be the circle $\mathbb{R}/2\pi\mathbb{Z}$ for continuous-state oscillators or the color wheel $\mathbb{Z}/\kappa\mathbb{Z}$, $\kappa \in \mathbb{N}$ for discrete-state oscillators. We designate a map $X : V \to \Omega$ to be a *phase configuration*, and say that if it takes a constant value across nodes (i.e., $X(v) = constant$ for all $v \in V$), then it is *synchronized*.

A *coupling* is a function $\mathscr{F}$ that maps each graph-dynamics pair (G, X_0) of graph and initial configuration $X_0 : V \to \Omega$ deterministically to a *trajectory* $(X_t)_{t\geq 0}$ of phase configurations $X_t : V \to \Omega$ from time/iterations 0 to T_0. In essence, $\mathscr{F}$ is the time evolution rule for a coupled oscillator model. The main problem we investigate in this work is stated as follows:

[Synchronization Prediction] Fix parameters $k \in \mathbb{N}$, $T_0 > 0$, and coupling $\mathscr{F}$. Given a k-node connected graph G with a Hamiltonian path P and trajectory $(X_t)_{0\leq t\leq T_0}$, predict the following indicator function

$$\mathbf{1}(X_t \text{ is eventually synchronized})$$

In this work, we tackle this synchronization problem by proposing a framework which not just maintains high accuracy over the theoretical baseline, but also place emphasis on *how* the model makes these predictions, which would enable us to learn new insights about the working of coupled oscillators and promote the use of machine learning based techniques for this field of work to expand to various potential applications.

Methods

Matrix factorization techniques provide a very powerful tool for one to describe various features of the data of interest in terms of a 'linear combination' of atomic elements.This problem has been studied over many decades under different names like factor analysis, topic modeling, etc. by various scientific fields [Berry and Browne (2005), Boutchko et al. (2015)]. The setup for the nonnegative matrix factorization is that we have a *data* matrix $X \in \mathbb{R}^{d \times n}$, and we seek the factorization of X into the product of a *dictionary* $W \in \mathbb{R}^{d \times R}$ and a *coding* $H \in \mathbb{R}^{R \times n}$, where the parameter R is the *rank* parameter that can be tuned based on how strong of a representation of our feature space we would like to achieve.

A *tensor* is a multi-dimensional array that is used to represent a generalized form of a matrix, and is suitable for representing multi-modal data – data that is being used to represent a complex set of features using more than just one form. Tensor factorization is thus a useful tool that can extract useful latent information out of data tensors. It comes in handy particularly so in our case, since we choose to represent our graph-dynamic pairs (G, X_0) as $\mathbf{X} = k \times k \times T$ 'color-coded' tensors. Conisder that the graph G has k nodes, and our dynamics X_0 have been studied from iterations 0 to T. Then, we use a tensor representational data-structure to encode these T slices of $k \times k$ adjacency matrices of the graph dynamics at each iteration, but replacing the non-zero entries $A[i, j]$ of the adjacency matrix with the difference of their phase dynamics at that time slice t as follows:

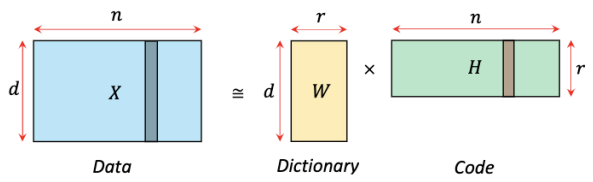

Figure 1: Nonnegative Matrix Factorization [Lyu et al. (2020)]

$$\mathbf{X}[i, j, t] := A_{ij} \min\left(X_t(i) - X_t(j), X_t(i) - X_t(j)\right) \pmod{\kappa}.$$

where $A = (A_{ij})$ is the adjacency matrix of G defined by $A_{ij} = \mathbf{1}(\{i, j\}) \in E)$. And this is the reason we dub this structure as the 'color-coded' tensor, as each of these tensors would contain multiple $2D$ matrix slices with different color encoding based on the phases of the nodes connected by that entry in the adjacency matrix.

These tensor factorization techniques can prove to be powerful tools for the task of synchronization prediction as they would enable us to succinctly capture the spatiotemporal features of our 'color-coded' tensor. The *spatial* aspect comes in because of our 'color-coded' slices of matrices, each of which contains spatial information about the how the coupled oscillator dynamics look at each time frame, and the *temporal* aspect comes in because of the third dimension bringing the notion of 'time' in understanding how these dynamics at every time frame are evolving into the next one, which is crucial to the model for it to predict synchronization. This spatiotemporal pattern is what makes our data multi-modal, and is the reason why we think the tensor decomposition techniques will work well for this task.

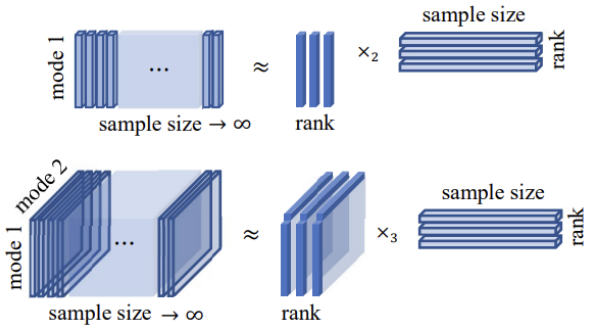

Figure 2: Online Matrix Factorization (Top) and Online CP Decomposition (Bottom) [Lyu et al. (2022)]

Finally, the pursuit of interpretability leads us to focusing on one of the standard tensor factorization paradigms, which is the `CANDECOMP/PARAFAC` (CP) decomposition [Tucker (1966), Harshman et al. (1970), Carroll and Chang (1970)]. In this setting, given a $n-$modal data tensor X, we seek to learn n matrices $W^{(1)}, \cdots, W^{(n)}$, each with R columns which is again the *rank* parameter, such that X is best approximated by the *sum of the outer products* of the respective columns of $U^{(i)}$'s [Lyu et al. (2022)].

We will consider a special case of CP-decomposition when the data tensor as well as all of its representational matrices are required to have nonnegative entries, since our entire tensor always only consists of nonnegative entries. Just like in the case of nonnegative matrix factorization work by Lee and Seung (1999), imposing this nonnegativity constraint in this tensor decomposition problem would enable us to learn interpretable features from our color-coded tensors as well. Finally, these learned features can be used to perform the classification task using simple techniques like logistic regression or more complicated techniques, as would be required.

Finally, using the online setting for applying these algorithms will help us eliminate the bottleneck of processing huge amounts of data at once or lose information by processing in batches by seeing examples 'on-the-go'. Online algorithms for these factorization methods have gained immense popularity in recent years [Guan et al. (2012), Rambhatla et al. (2019)] and we believe that this would be a great setting for the synchronization problem because we would not have to see the entire tensor at once, which could span 100s of iterations thus making our computational requirements simpler.

Timeline

The following is an outline of the timeline for completion of the project:

Timeline for the Project	
Timeline	**Task To Be Completed**
Summer 2023	Background Reading, Thorough Literature Review, Project Setup
15th October 2023	Apply Nonnegative Factorization Techniques
15th December 2023	Shift Focus to (Online) Tensor Factorization Techniques
15th February 2024	Run Final Experiments as Required, Compile and Interpret Results
15th April 2024	Wrap up the Project, Prepare for the Presentation, and Write Report
End of April 2024	Presentation at the Undergraduate Symposium 2024

I will meet with my project advisor Dr. Lyu either weekly or bi-weekly as required to report my findings, interpret results with him and brainstorm directions for further progress. This will also enable me to get regular feedback on my work.

Conclusion

In conclusion, we plan to use these tensor factorization techniques, with special emphasis on the nonnegative version of the CP-Decomposition to learn a strong feature representation of our 'color-coded' tensors which we believe would help us learn their spatiotemporal patterns in an interpretable manner. These features can then be used for the purpose of classification. To the best of our knowledge, this would be the first work of this kind to approach the prediction of synchronization of coupled oscillators using tensor factorization techniques, while also providing an interpretable framework to benefit not just the coupled oscillator and clock synchronization communities, but also enable this to be used widely to study various fields of interest in today's world.

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